Timing-Only Channel Estimation and Equalization for OFDM through Dispersive Time-Modulated RIS

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***Abstract*—A time-modulated reconfigurable intelligent surface (TM-RIS) that switches its elements within each OFDM symbol couples every subcarrier to the others through the harmonics of its control sequence, and a recent physically consistent model shows that this coupling is further shaped by the frequency-dispersive response of the elements. We give the first receiver for this channel. Regrouping the harmonic expansion by switching state, the coupling matrix is a sum of Toeplitz matrices fixed by the control timing, each weighted by an unknown input-side response that absorbs the element dispersion. This yields (i) a pilot-based low-rank estimator that needs no element model, only the control timing; (ii) blind recovery of an unknown RIS clock offset from the Bayesian evidence of the estimator; and (iii) exact matrix-free equalization. With two pilot symbols the receiver is within 0.1 bit/s/Hz of perfect channel knowledge at 20–30 dB, needs 16 times fewer pilots than a full basis-expansion estimator, decodes coded packets within 0.5 dB of perfect knowledge where the conventional receiver never decodes, and keeps 4–6 bit/s/Hz net of pilot overhead on 3GPP TDL channels up to 120 km/h.**

***Index Terms*—Reconfigurable intelligent surface, time modulation, OFDM, intercarrier interference, channel estimation.**

I. Introduction

Time-modulated (space-time-coding) metasurfaces switch their elements faster than the symbol rate and so radiate the impinging signal at harmonics of the switching frequency [2]. Aligned with OFDM symbols, time modulation is used for waveform security and directional modulation [3], [4]. At the legitimate receiver the harmonics couple subcarriers, and prior designs treat this coupling as interference or confine switching to guard intervals. Iudice *et al.* [1] recently gave a physically consistent model of a TM-RIS with frequency-dispersive (finite-memory) elements: the coupling is non-diagonal for any cyclic prefix, and receiver design was left open.

This letter designs that receiver. Its contributions are:

- a pilot-based estimator of the full coupling matrix that requires only the control timing, not the element model, by absorbing the element dispersion into the unknown input-side channel (a regrouping of [1, eq. (5)]);
- blind recovery of an unknown RIS clock offset, using the observation that an offset is exactly an opposite delay of the two hops, so that the pilot fit is nearly blind to it while the Bayesian evidence is not;
- an evaluation against the strongest baselines (full-harmonic basis expansion, literature equalizers), with coding, 3GPP channels, mobility, hardware impairments and pilot overhead.

We adopt, and do not claim, low-rank modelling of the cascaded channel, matrix-free CG equalization [5], the basis-expansion estimator [6] (as a baseline), the two-timescale learning of the static base-station–RIS hop [7], and evidence-based model selection for OFDM timing [8].

II. System Model

OFDM with M active subcarriers, spacing Δf and useful duration $T_u = 1/\Delta f$ passes through a RIS of Q elements. Element q follows a K-slot control sequence $s_q[k] \in \{1, 2\}$ of period T_u. In state s it has the single-pole response [1]

$$\Gamma_s(f) = -1 + \frac{2\xi_{r,s}}{j2\pi(f - f_{0,s}) + \xi_{r,s} + \xi_{i,s}}. \tag{1}$$

Under the generalized cyclic-prefix condition [1], $\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{w}$ with

$$H_{\bar{m},m} = \sum_{q=1}^{Q} A_q^{\mathrm{d}}(\bar{m})\, b_q^{[\bar{m}-m]}(f_m)\, A_q^{\mathrm{s}}(m), \tag{2}$$

where $A_q^{\mathrm{s}}, A_q^{\mathrm{d}}$ are the BS–element and element–user responses and $b_q^{[h]}(f)$ is the h-th harmonic of the switched element response.

III. Structure of the Coupling

Grouping the slot sum of [1, eq. (5)] by state gives

$$b_q^{[h]}(f) = \sum_{s=1}^{2} t_{q,s}[h]\, \Gamma_s(f), \tag{3}$$

$$t_{q,s}[h] = \frac{e^{-j\pi h/K}}{K} \operatorname{sinc}\frac{h}{K} \sum_{k:s_q[k]=s} e^{-j2\pi hk/K}. \tag{4}$$

Hence

$$\boldsymbol{H} = \sum_q \sum_s \boldsymbol{T}_{q,s} \circ \left(\boldsymbol{a}_q^{\mathrm{d}} (\boldsymbol{\Gamma}_s \odot \boldsymbol{a}_q^{\mathrm{s}})^{\top}\right), \tag{5}$$

with $\boldsymbol{T}_{q,s}$ Toeplitz and fixed by the timing alone. Three consequences drive the receiver. (a) The dispersion $\boldsymbol{\Gamma}_s$ multiplies the input side only, so it can be absorbed into an unknown per-state input response: no element model is needed. (b) Each $\boldsymbol{T}_{q,s}\boldsymbol{x}$ is a linear convolution, so $\boldsymbol{H}\boldsymbol{x}$ and $\boldsymbol{H}^{\mathsf{H}}\boldsymbol{y}$ cost $\mathcal{O}(QM \log M)$. (c) The envelope of $t_{q,s}[h]$ decays as $1/h$, so any truncated harmonic model, and any banded equalizer, leaves a residual floor.

IV. Receiver

A. Timing-only low-rank estimation

Per-hop responses are delay limited, so $\boldsymbol{a}_q^{\mathrm{d}} = \boldsymbol{F}\boldsymbol{u}_q$ and $\boldsymbol{\Gamma}_s \odot \boldsymbol{a}_q^{\mathrm{s}} = \boldsymbol{F}\boldsymbol{v}_{q,s}$ with a DPSS basis $\boldsymbol{F} \in \mathbb{C}^{M \times L}$. All elements see the same propagation paths, so the element-domain factors

share a rank-r subspace. Grouping elements with the same sequence into G groups,

$$\boldsymbol{H} = \sum_{g=1}^{G} \sum_{s=1}^{2} \boldsymbol{T}_{g,s} \circ \big((\boldsymbol{F}\boldsymbol{U})\boldsymbol{\Lambda}_g(\boldsymbol{F}\boldsymbol{V}_s)^{\top}\big), \tag{6}$$

with $\boldsymbol{U}, \boldsymbol{V}_s \in \mathbb{C}^{L\times r}$ and $\boldsymbol{\Lambda}_g \in \mathbb{C}^{r\times r}$. Identical sequences ($G = 1$) and cyclic shifts ($\boldsymbol{T}_{g,s} = \boldsymbol{D}_k \boldsymbol{T}_s \boldsymbol{D}_k^*$) are special cases. From N_p pilot symbols we estimate the factors by block alternating LMMSE with a DPSS-eigenvalue prior; each block is linear and the $\boldsymbol{\Lambda}$ step is convex. The initial point is the pooled linear LMMSE estimate of the per-state delay matrices. With two OFDM pilot symbols ($2M$ equations) the unknowns, $3Lr + Gr^2$, are few for compact delay windows. [TODO: state L, r used: $L = 16$, $r = 8$ at $M = 256$.]

B. Blind recovery of the RIS clock offset

A clock offset of d slots maps $\boldsymbol{T}_{g,s} \mapsto \boldsymbol{D}\boldsymbol{T}_{g,s}\boldsymbol{D}^*$ with $\boldsymbol{D} = \mathrm{diag}(e^{-j2\pi md/K})$, and

$$(\boldsymbol{D}\boldsymbol{T}\boldsymbol{D}^*) \circ (\boldsymbol{a}\boldsymbol{b}^{\top}) = \boldsymbol{T} \circ \big((\boldsymbol{D}\boldsymbol{a})(\boldsymbol{D}^*\boldsymbol{b})^{\top}\big) : \tag{7}$$

an offset is a delay of the user hop and an advance of the BS hop by dT_u/K. The DPSS basis still represents the shifted factors (projection error 10^{-4} in our setting), so the pilot residual is nearly flat in d, and a residual search fails at high SNR. The shifted factors are, however, improbable under the delay prior. We therefore choose d by maximizing the evidence of the linear first stage, $\boldsymbol{y} = \boldsymbol{\Phi}_d\boldsymbol{c} + \boldsymbol{w}$, $\boldsymbol{c} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{P})$:

$$\hat{d} = \arg\max_{d} \; -\boldsymbol{y}^{\mathsf{H}}\boldsymbol{C}_d^{-1}\boldsymbol{y} - \log\det \boldsymbol{C}_d, \tag{8}$$

$$\boldsymbol{C}_d = \boldsymbol{\Phi}_d\boldsymbol{P}\boldsymbol{\Phi}_d^{\mathsf{H}} + \sigma^2\boldsymbol{I}, \tag{9}$$

by a grid and golden-section search. Blind offset estimation has been studied for RIS backscatter (symbiotic radio) [9], where the RIS modulates its own data; here the offset is resolved jointly with the coupled channel.

C. Equalization

MMSE equalization by conjugate gradients on $(\hat{\boldsymbol{H}}^{\mathsf{H}}\hat{\boldsymbol{H}} + \sigma^2\boldsymbol{I})\boldsymbol{x} = \hat{\boldsymbol{H}}^{\mathsf{H}}\boldsymbol{y}$, with products through (5) as in [5]; here the representation is exact rather than a basis-expansion approximation. For $G = 1$ all elements share one Toeplitz kernel per state, so a product needs one batched FFT per state.

V. Numerical Results

OpenRIS n78 cell of [1, Table I], $f_c = 3.594\,\mathrm{GHz}$, $\Delta f = 30\,\mathrm{kHz}$, $Q = 16$, $K = 8$, duty 0.75, $M = 256$ unless stated; physical channels with shared paths (Rician 10 dB, 30 ns per hop) and 3GPP TDL-A/C/D [10]. Rates are MMSE achievable rates on the central half of the band.

Element model. Assuming the ideal ± 1 switch of earlier work costs 1.4/6.6 dB at 20/30 dB; a 10 MHz resonance error costs 0.7/3.4 dB; the timing-only estimator loses 0.08/0.11 dB.

Estimators. With two pilot symbols, ours reaches 6.52/9.68 bit/s/Hz at 20/30 dB against 6.61/9.92 with perfect knowledge, while a 25-harmonic basis-expansion estimator reaches 2.20/1.08 and the one-tap receiver 0.42/0.43 (Fig. 1).

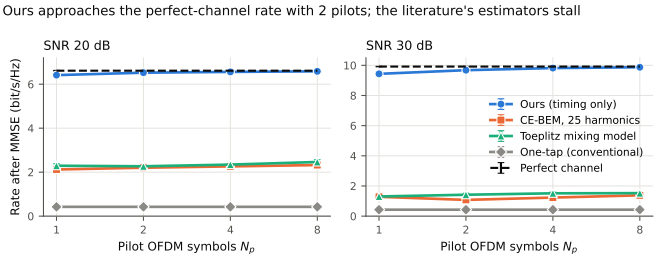


Fig. 1. Rate vs pilot symbols at 20 and 30 dB, 20 channels with 95% CIs.

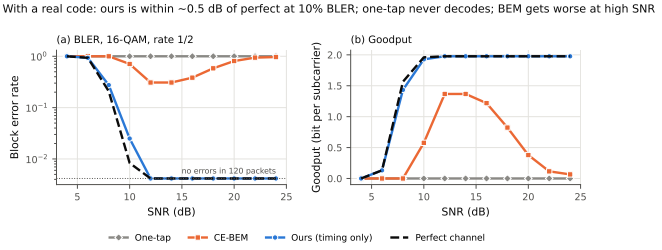


Fig. 2. Coded block error rate and goodput, rate-1/2 convolutional code, 16-QAM.

Keeping all 511 harmonics, the basis expansion needs 32 pilot symbols to reach 6.18/9.37: a 16-fold pilot saving. Banded equalizers fail even with the true channel (6.2 vs 9.9 bit/s/Hz at 30 dB with a ± 32 band), because of the $1/h$ tail; ten LSQR iterations match the full MMSE.

Full band. At $M = 3276$ (10 trials) the symbol error rate with two pilot symbols is 1.09× that of perfect knowledge at 20 dB/16-QAM and 1.54× at 28 dB/64-QAM (1.05× and 1.14× after one decision-directed pass); the one-tap receiver misses 84–96% of symbols.

Coding. With a rate-1/2 convolutional code and 16-QAM, ours has block error rate 0.025 at 10 dB and none from 12 dB, within 0.5 dB of perfect knowledge; the one-tap receiver never decodes and the basis expansion floors at 0.31 (Fig. 2).

Timing. An unknown off-grid clock offset is found to 5 ns rms from two pilot symbols; at 30 dB the loss is 0.40 dB against 0.41 dB with the offset known, while a residual search loses 4.8 dB (Fig. 3). Random per-edge jitter is not absorbed and must stay below about 4 ns at 30 dB.

Channels and pilot overhead. TDL channels with 100 ns delay spread need eight pilot symbols for a gap of 0.08–0.24 bit/s/Hz at 20 dB. Net of pilot overhead, with the refresh interval optimized and common-phase tracking as in 5G, ours keeps 5.1–5.7, 4.7–5.1 and 3.9–4.3 bit/s/Hz on TDL-A/C/D at 3, 30 and 120 km/h (overheads 5–7%, 5–10% and 12%), against 1.5–3.7 for the best basis expansion (Fig. 4).

Complexity. In MATLAB at $M = 3276$ the batched CG equalizer takes 1.1 s per 12 symbols against 3.0 s for the dense solve, which no longer fits in memory at $M = 6552$; warm-started channel updates are 8.6× faster than estimation from scratch.

Limitations. Results are simulated with one published element model and no mutual coupling. At 120 km/h a pilot is needed every ∼8 symbols; two tracking receivers did not

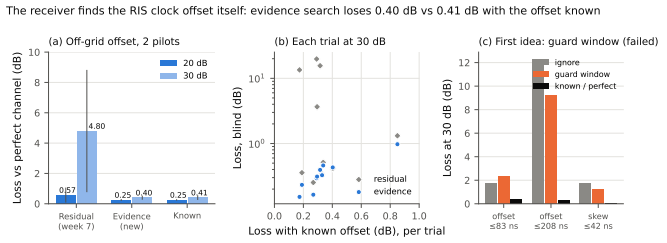


Fig. 3. Blind recovery of an off-grid RIS clock offset from two pilot symbols.

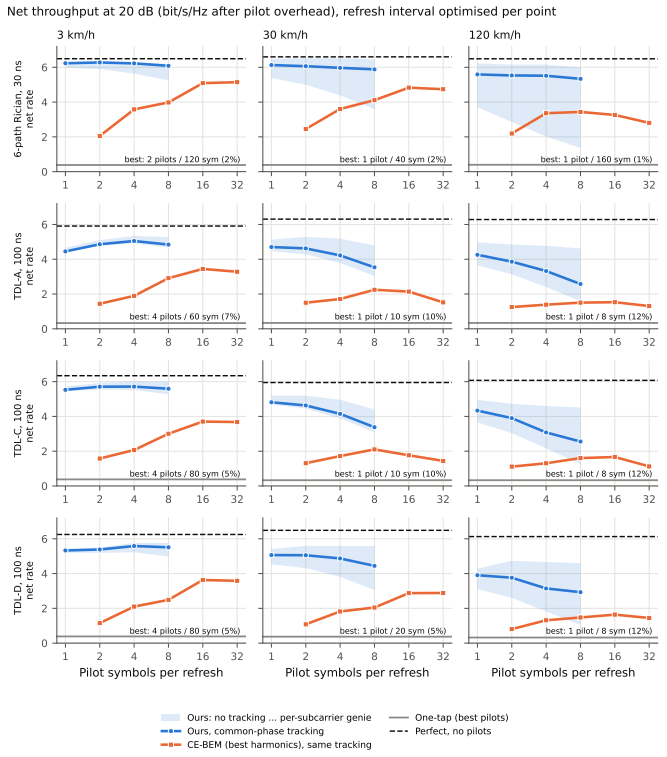


Fig. 4. Net throughput after pilot overhead vs pilot symbols per refresh, 20 dB. [TODO: a two-row compact version for the letter (6-path and TDL-C).]

improve on this, and at 30 dB the Doppler spread within a symbol, not modelled here, limits every receiver. At 300 ns delay spread the static BS–RIS hop must be learned over $\sim$100 frames (TDL-A/C, gap 0.23–0.26 bit/s/Hz at 20 dB); channels whose taps exceed the cyclic prefix are out of scope.

VI. Conclusion

Grouping the TM-RIS harmonic expansion by state separates the control timing, known at the receiver, from the element dispersion, which can be absorbed into the unknown channel. A receiver built on this separation needs no element model, finds the RIS clock offset by itself, equalizes without forming the coupling matrix, and turns the harmonic coupling that conventional receivers suffer from into usable rate.

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